

Faculty of Engineering

Department of Management Science and Engineering

MSE 333 Fall 2026: Simulation Analysis and Design

Instructor	Hassan Shavandi
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Office hours	Wed 2:30pm to 3pm or by appointment Office: CPH 4311

Teaching Assistants		
Name	email	Responsibility
Omar Hussein	omahussein@uwaterloo.ca	Tutorial / Grading
Kaan Apak	kapak@uwaterloo.ca	Lab / Grading

	Time	Place
Lecture	M/W 1:00pm to 2:20pm	CPH 3681
Tutorial	Tuesday: 11:30am – 12:20pm	DWE 2527
Lab	Friday: 11:30am – 12:50pm	CPH 1346

Course Description

This course introduces the use of discrete event simulation as an approach for understanding and analyzing complex management systems. Topics include an introduction to simulation modelling, general purpose and special purpose simulation languages, designing valid and credible simulation models, input data analysis, and output performance analysis.

Prerequisites: Calculus, Probability and Statistics.

Simulation program: ARENA, Simpy, and Microsoft Excel.

Textbooks:

Banks, J., Carson II, J.S., Nelson, B.L., and Nicol, D.M. (2010),
Discrete-Event System Simulation, 5th Edition, Prentice Hall, New York.
ISBN-13: 978-0-13-606212-7 ISBN-10: 0-13-606212-1

Kelton M. David, & et al. (2024).

Simulation with ARENA. 7th Edition, Mc Graw Hill Education.

ISBN-978-1-266-27572-2

Course Materials

1. UW-Lean: lecture notes, lab notes, and tutorial notes. Assignments, and other supplementary materials.
2. Textbooks

Intended Learning Outcomes

By the end of the course, students should be able to:

1. Define key terms and concepts in simulation analysis and design [1]*
2. Describe the simulation modeling process and methods [1]*
3. Use statistical methods to find the type of distributions, estimate the parameters, and implement goodness-of-fit tests. [3, 5]*
4. Analyze, design and build simulation models for stochastic systems. [2, 3, 4, 5, 6, 7, 8, 12]*
5. Verify and validate simulation models using statistical methods and software (e.g., Excel and Arena). [3, 5, 12]*
6. Investigate the system's performance measures through the simulation model for different scenarios and conclude the results. [3, 5, 12]*
7. Be familiar with emerging trends in simulation: Digital Twins, Integration of Simulation and ML, and Simulation Optimization. [2, 4, 5]*

* *Graduate Attributes required for Accreditation, see appendix.*

General Course Instructions

- **Lectures** will be used to introduce simulation modelling techniques to model real-life systems, and methods to analyze those models.
- **Lab sessions:** Concepts discussed in the lectures will be applied to develop working simulation models using ARENA and Microsoft Excel. Often new modelling and analytical techniques will be introduced in Lab sessions.
- **Tutorials:** Objective is to illustrate the concepts discussed in the lecture through examples and to answer student's queries. However, new, or supplementary materials may be discussed in tutorials if necessary.
- It is highly recommended to attend all sessions. Assignments, projects, and exams will require an in-depth understanding of the material discussed in lectures, labs and tutorials.

- Group discussion of assignment problems is encouraged. However, every student must complete his/her assignments independently.
- All assignments and projects must be submitted online via **Crowdmark** and/or **UW-learn Dropbox** before the deadline. Submissions on Learn without Crowdmark submission will not be considered for marking. Late submissions will not be accepted unless a strong reason is presented.
- An acceptable project, whether late or not, must be submitted to pass the course.
- All announcements related to the course will be made through UW-learn, email, or in class.
- Your questions, comments and suggestions will be greatly appreciated. Please use the discussion forums or email instructor/TAs.

Marking Scheme:

Description	%	Comment
Assignments	28%	4 assignments (20%): Individual work Lab activities (8%): Group work
Project	18%	Can be done individually or group of 2
Midterm exam	18%	Lectures 1, 2, 3, 4
Classworks and Pop quizzes	8%	4% Bonus mark (Every session may have a classwork or pop-quiz)
Final Exam	32%	All chapters

Assignments:

Assignment	Mark
1	6%
2	6%
3	4%
4	4%

MSE 333, Fall 2026 Course Schedule (note: Date of topics are approximate)

Week of	Theme	Topics	Assignments/Evaluation
W1: Sep 9	Lec	Lec1. Introduction to Simulation	
	Lab	No lab	
	Tut	No tutorial	
W2: Sep 14 & 16	Lec	Lec 1. Introduction to Simulation and Lec 2. Simulation in Excel	Enrolling project groups Due date: Sep 18, 11:59pm
	Lab		
	Tut		
W3: Sep 21 & 23	Lec	Lec 2. Simulation in Excel	
	Lab		
	Tut		
W4: Sep 28 & 30	Lec	Lec 2. Simulation in Excel and Lec 3. Simulation principles	Assignment 1 (due: Oct 2, 11:59pm): Lec 1 & 2
	Lab		
	Tut		
W5: Oct 5 & 7	Lec	Lec 3. Simulation principles	
	Lab		
	Tut		
W6: Oct 12 & 14	Reading Week		
W7: Oct 19 & 21	Lec	Lec 4. Queuing Models	Assignment 2 (due: Oct 23, 11:59pm): Lec 3 & 4
	Lab		
	Tut		
W8: Oct 26 & 28	Lec	Lec 5. Random Numbers	Midterm Exam (Lec 1, 2, 3, and 4) Oct 26, 1:00pm to 2:20pm RCH 301
	Lab		
	Tut		
W9: Nov 2 & 4	Lec	Lec 5 & 6. Random Numbers and Random Variates	
	Lab		
	Tut		
W10: Nov 9 & 11	Lec	Lec 7. Input Modeling: Statistical distribution fitting	Assignment 3 (due: Nov 13, 11:59pm): Lec 5, 6, 7
	Lab		
	Tut		
W11: Nov 16 & 18	Lec	Lec 8. Verification and Validation	
	Lab		
	Tut		
W12: Nov 23 & 25	Lec	Lec 9. Estimation of Absolute and Relative Performances	Assignment 4 (Nov 27, 11:59pm): Lec 8, 9
	Lab		
	Tut		
W13: Nov 30 & Dec 2	Lec	Lec 10. Emerging Trends in Simulation: Digital Twins, Integration of Simulation and ML, and Simulation Optimization	
	Lab		
	Tut		
W14: Dec 7	Lec	Q&A and Final exam review	Project report Due: Dec 6, 11:59pm
Final exam			TBS by Registrar

Guidelines for Assignments submission

Unless otherwise specified, all assignments must be submitted online in two parts:

Answers File (pdf) in Crowdmark and **Program File(s) (if applicable) in Learn Dropbox folder.**

In case of using any application like Excel or Arena, a screenshot of the pages should be included in the PDF file and then the application files should be submitted through created Learn Dropbox folder.

Note: In case of simulation system design problem, an outline of the program or code (pseudo-code and/or flowchart) must be included in the pdf file posted on Crowdmark. Pdf file should be self-explanatory without your program file.

Program File: ARENA or Excel maybe used to develop the simulation models. Program files must be submitted to have your problems marked unless otherwise specified.

- Please name your program files appropriately and submit them separately uncompressed.
Eg: "MSci333_StudentID_Asg*_P**"
(*Assignment Number; ** Problem Number)
- *Excel:* Label the columns and make comments appropriately.
- *ARENA:* Name the entities, variables, attributes and modules appropriately and in a self-explanatory manner.

Note: name the assignment file and program file(s) appropriately in the following format:

MSE 333_Student ID_Asgn#_Prob#

- Asgn#: Assignment number
- Problem#: Problem number

University Policies

Academic Integrity: In order to maintain a culture of academic integrity, members of the University of Waterloo are expected to promote honesty, trust, fairness, respect and responsibility. (Check www.uwaterloo.ca/academicintegrity for more information.)

Discipline: A student is expected to know what constitutes academic integrity, to avoid committing academic offences, and to take responsibility for his/her actions. A student who is unsure whether an action constitutes an offence, or who needs help in learning how to avoid offences (e.g., plagiarism, cheating) or about "rules" for group work/collaboration should seek guidance from the course professor, academic advisor, or the Undergraduate Associate Dean. When misconduct has been found to have occurred, disciplinary penalties will be imposed under [Policy 71](#) – Student Discipline. For information on categories of offenses and types of penalties, students should refer to [Policy 71](#) - Student Discipline.

Grievance: A student who believes that a decision affecting some aspect of his/her university life has been unfair or unreasonable may have grounds for initiating a grievance. Read [Policy 70](#) - Student Petitions and Grievances, Section 4.

Appeals: A student may appeal the finding and/or penalty in a decision made under [Policy 71](#) - Student Petitions and Grievances (other than regarding a petition) or [Policy 71](#) – Student Discipline if a ground for an appeal can be established. Read [Policy 72](#) - Student Appeals.

Turnitin: Text matching software (Turnitin®) will be used to grade and screen assignments in this course. Students' submissions are stored on a U.S. server, therefore students must be given an alternative, if they are concerned about their privacy and/or security. Students will be given due notice, in the first week of the term and/or at the time assignment details are provided, about arrangements and alternatives for the use of Turnitin in this course. It is the responsibility of the student to notify the instructor if they, in the first week of term or at the time assignment details are provided, wish to submit alternate assignment.

Note for students with disabilities: The Accessibility Services office, located in Needles Hall Room 1401, collaborates with all academic departments to arrange appropriate accommodations for students with disabilities without compromising the academic integrity of the curriculum. If you require academic accommodations to lessen the impact of your disability, please register with the AS office at the beginning of each academic term.

Appendix

All engineering programs are reviewed by the Canadian Engineering Accreditation Board (CEAB). One of the required accreditation criteria is that institutions ensure students have sufficient knowledge and proficiency with respect to the 12 Graduate Attributes (GAs) listed below. These attributes are mapped to the learning objectives in each course for assessment, as shown in the brackets. This allows the program to both comply with CEAB requirements and continuously improve.

Graduate Attributes

#	Attribute Name	Attribute Definition
1	Knowledge Base	Demonstrated competence in university level mathematics, natural sciences, engineering fundamentals, and specialized engineering knowledge appropriate to the program.
2	Problem Analysis	An ability to use appropriate knowledge and skills to identify, formulate, analyze, and solve complex engineering problems in order to reach substantiated conclusions.
3	Investigation	An ability to conduct investigations of complex problems by methods that include appropriate experiments, analysis and interpretation of data, and synthesis of information in order to reach valid conclusions.
4	Design	An ability to design solutions for complex, open-ended engineering problems and to design systems, components or processes that meet specified needs with appropriate attention to health and safety risks, applicable standards, and economic, environmental, cultural and societal considerations.
5	Use of Engineering Tools	An ability to create, select, apply, adapt, and extend appropriate techniques, resources, and modern engineering tools to a range of engineering activities, from simple to complex, with an understanding of the associated limitations.
6	Individual and Teamwork	An ability to work effectively as a member and leader in teams, preferably in a multi-disciplinary setting.

7	Communication Skills	An ability to communicate complex engineering concepts within the profession and with society at large. Such ability includes reading, writing, speaking and listening, and the ability to comprehend and write effective reports and design documentation, and to give and effectively respond to clear instructions.
8	Professionalism	An understanding of the roles and responsibilities of the professional engineer in society, especially the primary role of protection of the public and the public interest.
9	Impact on Society	An ability to analyze social and environmental aspects of engineering activities. Such ability includes an understanding of the interactions that engineering has with the economic, social, health, safety, legal, and cultural aspects of society, the uncertainties in the prediction of such interactions; and the concepts of sustainable design and development and environmental stewardship.
10	Ethics and Equity	An ability to apply professional ethics, accountability, and equity.
11	Economics and Project Management	An ability to appropriately incorporate economics and business practices including project, risk, and change management into the practice of engineering and to understand their limitations.
12	Life-long Learning	An ability to identify and to address their own educational needs in a changing world in ways sufficient to maintain their competence and to allow them to contribute to the advancement of knowledge.